

HW1 (Titanic Dataset)

Tsung-Yu Lee

2024-03-06

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Summary statistics

Load the titanic_train dataset in r package "titanic".

```
library(titanic)
titanic_train[titanic_train == ""] <- NA
```

Explore the titanic_train dataset by using latex(describe()) function,

```
library(Hmisc)
latex(describe(titanic_train), file = "", caption.placement = "top")
```

titanic_train												
12 Variables						891 Observations						
PassengerId												
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
891	0	891	1	446	297.3	45.5	90.0	223.5	446.0	668.5	802.0	846.5
lowest : 1 2 3 4 5, highest: 887 888 889 890 891												

Survived

n	missing	distinct	Info	Sum	Mean	Gmd
891	0	2	0.71	342	0.3838	0.4735

Pclass

n	missing	distinct	Info	Mean	Gmd
891	0	3	0.81	2.309	0.8631

Value	1	2	3
Frequency	216	184	491
Proportion	0.242	0.207	0.551

For the frequency table, variable is rounded to the nearest 0

Name

n	missing	distinct
891	0	891

lowest : Abbing, Mr. Anthony
highest: Yousseff, Mr. Gerious

Abbott, Mr. Rossmore Edward
Yrois, Miss. Henriette ("Mrs Harbeck")

Abbott, Mrs. Stanton (Rosa Hunt)
Zabour, Miss. Hileni

Sex

n	missing	distinct
891	0	2

Value	female	male
Frequency	314	577
Proportion	0.352	0.648

Age

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
714	177	88	0.999	29.7	16.21	4.00	14.00	20.12	28.00	38.00	50.00	56.00

lowest : 0.42 0.67 0.75 0.83 0.92, highest: 70 70.5 71 74 80

SibSp

n	missing	distinct	Info	Mean	Gmd
891	0	7	0.669	0.523	0.823

Value	0	1	2	3	4	5	8
Frequency	608	209	28	16	18	5	7
Proportion	0.682	0.235	0.031	0.018	0.020	0.006	0.008

For the frequency table, variable is rounded to the nearest 0

Parch

n	missing	distinct	Info	Mean	Gmd
891	0	7	0.556	0.3816	0.6259

Value	0	1	2	3	4	5	6
Frequency	678	118	80	5	4	5	1
Proportion	0.761	0.132	0.090	0.006	0.004	0.006	0.001

For the frequency table, variable is rounded to the nearest 0

Ticket

n	missing	distinct
891	0	681

lowest :	110152	110413	110465	110564	110813
highest:	W./C. 6608	W./C. 6609	W.E.P. 5734	W/C 14208	WE/P 5735

Fare

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
891	0	248	1	32.2	36.78	7.225	7.550	7.910	14.454	31.000	77.958	112.079

lowest : 0 4.0125 5 6.2375 6.4375 , highest: 227.525 247.521 262.375 263 512.329

Cabin

n	missing	distinct
204	687	147

lowest : A10 A14 A16 A19 A20, highest: F33 F38 F4 G6 T

Embarked

n	missing	distinct
889	2	3

Value	C	Q	S
Frequency	168	77	644
Proportion	0.189	0.087	0.724

Table 1

Display some table 1s' by variable "Survived".

```
library(table1)
table1(~ Pclass+Sex+Age+SibSp+Parch+Fare+Embarked|Survived, data = titanic_train)
```

	0	1	Overall
	(N=549)	(N=342)	(N=891)
Pclass			
Mean (SD)	2.53 (0.736)	1.95 (0.863)	2.31 (0.836)
Median [Min, Max]	3.00 [1.00, 3.00]	2.00 [1.00, 3.00]	3.00 [1.00, 3.00]
Sex			
female	81 (14.8%)	233 (68.1%)	314 (35.2%)
male	468 (85.2%)	109 (31.9%)	577 (64.8%)
Age			
Mean (SD)	30.6 (14.2)	28.3 (15.0)	29.7 (14.5)
Median [Min, Max]	28.0 [1.00, 74.0]	28.0 [0.420, 80.0]	28.0 [0.420, 80.0]
Missing	125 (22.8%)	52 (15.2%)	177 (19.9%)
SibSp			
Mean (SD)	0.554 (1.29)	0.474 (0.709)	0.523 (1.10)
Median [Min, Max]	0 [0, 8.00]	0 [0, 4.00]	0 [0, 8.00]
Parch			
Mean (SD)	0.330 (0.823)	0.465 (0.772)	0.382 (0.806)
Median [Min, Max]	0 [0, 6.00]	0 [0, 5.00]	0 [0, 6.00]
Fare			
Mean (SD)	22.1 (31.4)	48.4 (66.6)	32.2 (49.7)
Median [Min, Max]	10.5 [0, 263]	26.0 [0, 512]	14.5 [0, 512]
Embarked			
C	75 (13.7%)	93 (27.2%)	168 (18.9%)
Q	47 (8.6%)	30 (8.8%)	77 (8.6%)
S	427 (77.8%)	217 (63.5%)	644 (72.3%)
Missing	0 (0%)	2 (0.6%)	2 (0.2%)

Missing Pattern

Display missing patterns of each variable in titanic_train dataset.

```
library(DataExplorer)
plot_missing(titanic_train)
```

